

Mnemosyne — Execution Plan B: Benchmarking & the Leaderboard

Goal: prove Mnemosyne’s results credibly, and build the field’s neutral memory-benchmark leaderboard

Version: 1.0 · **Date:** 2026-07-08 · **Status:** Proposed (awaiting go) **Scope:** the public benchmark harness (`eval/public/`), the publication protocol, and the greenfield `leaderboard/` + `web/` products. **Companion doc:** *Execution Plan A — The Memory System* builds the capabilities this plan measures. This plan owns measurement, publication, and the leaderboard. Where Plan A says “measured/published,” the authority is here. **Audience:** an autonomous engineering agent (or fleet) executing end-to-end, plus human operators for governance and third-party reproduction.

0. How to use this document (executing agent, read first)

Execution spec, not prose. Two parts: **Part I** benchmarks Mnemosyne credibly; **Part II** builds the neutral leaderboard. Part I’s harness is reused as Part II’s engine. Every task has a stable ID (`M1.2` , `L0.1`), a **DoD**, and required **artifacts**.

Hard guardrails (violating any = stop and escalate):

- 1. **PBPP governs every public number** (§2). No public claim without the full artifact bundle (§M2) and independent reproduction (§M3).
- 2. **Retrieval-recall and LLM-judged-QA are never blended**. Separate columns, always. Disclose judge model + prompt. (This is the single most common field failure — MemPalace, the mem0↔Zep dispute.)
- 3. **Never tune on a held-out/test split** — our own system included. Contamination discipline (§M1, §L1) applies to us exactly as to submitters.
- 4. **No self-defined benchmark is ever a headline claim** (the gbrain “BrainBench” anti-pattern). Mnemosyne’s private suite (0.977 / 0.983 / ECE 0.0063 / poison 1.0 / 149.5 ms) stays internal QA.
- 5. **Neutrality is structural, not asserted** (§L0). If we operate the leaderboard *and* compete, the firewall in §L0 is mandatory, or the board is not credible.

Ground-truth references (source of truth over this doc if they conflict): `docs/research/AI-Memory-Systems-Market-Research-2026.md` (the competitive + benchmark landscape and why LoCoMo is contested), `docs/blueprint/Mnemosyne-Performance-and-Refactoring-Blueprint.md` (§9.2.7 slates LongMemEval/BEAM wiring and carries the current benchmark-posture/honesty rule), `eval/README.md` , `eval/provider_bakeoff/README.md` .

DoD template: *Harness/site code merged + artifacts written to the named path + a one-paragraph result note in `leaderboard/reports/` + public data-repo updated (Part II) + independent reproduction on file (headline numbers).*

1. Objectives & success criteria (measurable)

Part I — Benchmark Mnemosyne credibly

#	Win condition	Definition
M-i	Public benchmarks wired	LongMemEval (retrieval), HippoRAG multi-hop, MemoryAgentBench, BEAM runnable in <code>eval/public/</code> with artifact bundles
M-ii	Metric families separated	Every result reports deterministic-retrieval and LLM-judged-QA in distinct columns
M-iii	Reproducible	Every headline number reproduced by an independent third party from the bundle alone
M-iv	Honest posture	Charter upgraded to PBPP; private suite never headlined

Part II — The neutral leaderboard

#	Win condition	Definition
L-i	It exists & is trusted	Public site + open harness + methods paper + openly-licensed results dump
L-ii	Structurally neutral	Operator runs every system under one harness; hard firewall on our own entry (§L0)
L-iii	Resists gaming	Hidden/rotated split, contamination controls, signed build==public attestation, honest LLM-judge reporting
L-iv	Becomes the default	Multi-track (conversational / agentic / multi-session); ≥8 systems live; academic co-sign

Explicit non-goals / anti-patterns

- No headlining **LoCoMo** or **DMR/MSc** (contested / saturated — market research §6).
- No presenting **recall@k as QA accuracy** (the MemPalace failure).
- No `top_k = entire-candidate-pool` “100%” retrieval theater.
- No “neutral” board that quietly advantages our own entry (the LMArena “Leaderboard Illusion”).

2. Strategic framing — the two hard problems, resolved

Problem 1 — the honesty-charter tension. The perf blueprint §9.2.7 and `eval/provider_bakeoff/README.md` currently forbid citing public benchmarks in headline claims — correct *while none had been run under our discipline*. Resolution: upgrade the prohibition into a **Public-Benchmark Publication Protocol (PBPP)**, stricter than any competitor’s practice:

PBPP. A public number may be published only if (a) produced by the pinned public harness in `eval/public/`, (b) the full artifact bundle (§M2) is released simultaneously, (c) retrieval-recall and LLM-judged-QA are reported in separate columns with judge model + prompt disclosed, (d) the private golden suite is never conflated with it, and (e) an independent third party reproduces it from the bundle. Private-suite numbers remain internal QA and are never headline public claims.

This turns honesty into the moat: our published numbers become the most reproducible in a field full of contested vendor claims. **Task M4:** update `docs/blueprint/...§9.2.7` and `eval/provider_bakeoff/README.md` to reference PBPP.

Problem 2 — the neutrality paradox. We cannot be referee and champion on trust alone; vendor-run-where-they-win boards get dismissed (OmniMemEval, omegamax, the LMArena “Illusion”). Resolution (§L0): the leaderboard runs under **independent governance with a hard firewall**; the operator **runs every system itself under one identical harness** (killing the “you misconfigured us” defense that defined the mem0↔Zep dispute); Mnemosyne is entered and scored **by the same rules as everyone else**; all raw artifacts are public. We win by having the best *reproducible* numbers on neutral turf — not by controlling the scoreboard.

PART I — Benchmark Mnemosyne credibly

M1 — Wire public benchmarks into `eval/public/` (deterministic-first) — *Wave E kickoff*

Blueprint §9.2.7 slates this. Build as a new isolated tree so the private suite stays clean; drive only the public `mneme` CLI (same rule as `eval/run_eval.py`).

- **M1.1 Harness scaffold.** `eval/public/` runner: pins each benchmark to a commit, emits the §M2 bundle. DoD: `mneme eval-public --suite X` runs end-to-end and writes traces.
- **M1.2 LongMemEval — retrieval-recall track first.** Deterministic Recall@k / nDCG using the dataset’s session/turn gold labels; **no LLM in scoring**. DoD: R@5 with Wilson CI + per-question retrieval traces. (Targets Plan A S-i.)
- **M1.3 HippoRAG multi-hop suite.** MuSiQue / 2WikiMultiHopQA / HotpotQA — recall@2/@5 + EM/F1, fully deterministic; exercises the graph+PPR channel. DoD: table vs published HippoRAG 2 baselines. (Plan A S-ii.)
- **M1.4 MemoryAgentBench adapter (upstream PR).** Mostly deterministic (SubEM/exact-match); contribute a Mnemosyne adapter upstream — the closest thing to an “official submission.” DoD: PR open + local results. (Exercises conflict-resolution / belief-revision.)
- **M1.5 BEAM (prestige, LLM-judged).** Disclosed reader model; publish as “Mnemosyne + .” DoD: BEAM-1M result + full config disclosure. (Plan A S-iii.)
- **M1.6 Scheduled CI job.** Runs deterministic public suites on a cadence; regression alerts only; never tune-to-test.

M2 — Reproducibility artifact bundle (the PBPP standard)

Every public result ships: pinned harness commit + `uv / pip` runner; per-question traces (**what was stored, what was retrieved, final answer**); disclosed judge model + prompt + all configs; system build fingerprint (we already emit `sha256:...` release fingerprints); Wilson/bootstrap CIs; one-command reproduce script. **DoD:** an independent *agent* reproduces the number from the bundle alone.

M3 — Third-party reproduction

Commission a genuine external party to reproduce every headline public number from the bundle before any public claim. **DoD:** signed reproduction note in `leaderboard/reports/`.

M4 — Charter → PBPP

Update blueprint §9.2.7 and the provider-bakeoff README to reference PBPP (§2). **DoD:** docs updated; CI lint points contributors to PBPP.

Benchmark slate (deterministic-first)

Tier	Benchmark	Track	Scoring	Why
Primary	LongMemEval (retrieval-recall)	conversational	Deterministic	Publishable recall@k; the flag everyone plants
Primary	HippoRAG suite (MuSiQue/2Wiki/HotpotQA)	multi-hop	Deterministic (R@k, EM/F1)	Cleanest deterministic; exercises our PPR channel
Strong	MemoryAgentBench	agentic/conflict	Mostly deterministic	Upstream adapter = closest to an “official” submission
Prestige	BEAM (1M/10M tok)	long-horizon	LLM-judged (disclosed reader)	Un-saturated; differentiating at scale
Utility	STATE-Bench (Microsoft)	enterprise task success	Deterministic task-completion	Shows memory improves real tasks
Internal-only	private v2 suite, LongMemEval-QA	—	private / LLM-judged	QA + regression signal; never a headline claim
Avoid headlining	LoCoMo, DMR/MSC	—	contested / saturated	Run silently at most; never lead with them

Part I exit gate: M-i...M-iv met; PBPP in force; third-party reproduction on file.

PART II — The neutral Memory Leaderboard

Working name: **OpenMemBench** (final naming is a branding decision; recommendation: a neutral, non-Mnemosyne-branded identity to preserve independence).

L0 — Governance & neutrality FIRST (the credibility moat — build before any code)

- **L0.1 Independent governance charter.** Multi-institution advisory board (invite ≥2 academics who authored memory benchmarks); published, dated COI policy; the operator’s own entry (Mnemosyne) gets **no** extra runs, no private tuning window, no earlier held-out access, no score-retraction rights. Model on HELM’s disclosed-funding posture + LMArena’s post-“Illusion” reforms.
- **L0.2 One-harness rule.** Operator runs **every** system itself under one identical harness (the reused `eval/public/` core); vendor-submitted numbers are never accepted as-is. Structurally eliminates the “you misconfigured us” dispute.
- **L0.3 Survivability.** Openly-licensed results dump with durable, citable URLs + non-single-owner hosting so the board outlives its host (the Papers-with-Code lesson).
- **L0.4 Methodology paper.** Publish a peer-review-grade methods paper at launch (arXiv 2603.07670 explicitly calls for a “GLUE-style shared leaderboard for agent memory” — that gap is the opening).

L1 — Benchmark methodology & harness (requirements checklist)

Each item maps to a proven precedent (full source map in Appendix B): - **L1.a Multi-track, GLUE-style:** conversational · agentic · multi-session, shared metric stack. - **L1.b Separate metric families:** deterministic retrieval (recall@k, EM, programmatic assertions) vs LLM-judged QA — never blended. - **L1.c Contamination controls:** machine-validated training-data disclosure (MTEB `training_datasets`); surface a **zero-shot overlap** score *without hiding honest disclosers*; fresh-vs-circulated probes (SWE-bench-Illusion method). - **L1.d Hidden/held-out + rotation:** publish train/dev, keep a **private scoring split** (ARC-AGI); public/private split with periodic reveal (Kaggle); scheduled task rotation + human baseline per track (GLUE saturation lesson). - **L1.e Anti-gaming:** signed **attestation that evaluated build == public release** (LMArena’s Llama-4 fix); submission throttling (SuperGLUE 2/day); disclose *all* variants tested; public retirement list. - **L1.f Keep the judge honest:** publish the judge’s acceptance rate on intentionally-wrong-but-topical answers (beat LoCoMo’s 62.8%); multi-judge averaging; rotating human adjudication; prefer deterministic grading where possible. - **L1.g Systems-fair reporting:** efficiency (latency + cost/query) first-class, budget-normalized; CIs + rank ranges with overlapping-CI systems treated as **ties**; set-independent aggregation (Borda, not mean-win-rate). - **L1.h Reproducibility:** pinned harness commit, full per-question traces public, open-source-to-rank.

L2 — Website architecture & build (greenfield `web/` + `Leaderboard/`)

- **Stack (recommendation):** static-first for durability + trust — **Astro or Next.js (static export)** front-end; results stored as **flat, versioned JSON/Parquet in a public git repo** (not a hidden DB) so every number is auditable and URLs are permanent; light charting lib; a **per-question trace browser** (HELM’s “browse all predictions” pattern). CDN-hosted; mirrored data dump.
- **Data model:** `system → track → benchmark_version → run(commit, build_fingerprint, config, judge) → per_question_traces → metrics(with CIs)`.
- **Submission flow:** reviewed **PR with provenance link** (MTEB model); CI validates schema + build attestation; organizer runs the eval; results + traces published.
- **DoD:** site renders the leaderboard from the public data repo; ≥1 system fully browsable end-to-end (traces included).

L3 — Content / explainer layer (“explain memory to everyone”)

Plain-language explainers of each major system’s architecture (reuse `docs/research/AI-Memory-Systems-Market-Research-2026.md`), the **recall-vs-QA** distinction front-and-center, and a transparent methodology page. **DoD**: explainer pages published; reviewed for accuracy against primary sources.

L4 — Launch & community

Seed by running **all major systems** (mem0, Zep/Graphiti, Letta, Cognee, MemOS, supermemory, HippoRAG, + Mnemosyne) under the one harness; publish the methods paper; open submissions + a public audit/dispute channel; **release raw head-to-head data** (LMArena’s single most trust-restoring act). **DoD**: public launch with ≥8 systems and a reproducible methods paper.

Part II exit gate: L-i...L-iv met; independent board seated; ≥8 systems live; methods paper public; Mnemosyne entered under identical rules.

5. Cross-cutting: anti-gaming, verification, risks

Failure modes to design against (real 2026 episodes): - MemPalace: recall@k reported as QA; `top_k = whole pool`; hand-patching dev questions. → Enforced by L1.b, L1.d, per-question trace publication. - gbrain: self-defined “BrainBench” headline. → Forbidden by guardrail #4 / PBPP. - LMArena “Leaderboard Illusion”: operator/unequal access. → Neutralized by L0.1–L0.2. - LoCoMo: 6.4% wrong answer key + lenient judge. → We don’t headline LoCoMo; judge honesty published (L1.f).

Verification: every headline public number reproduced by an independent party before publication (M3); every leaderboard result reproducible from its bundle; a standing “red-team the number” review before any external claim.

Risk register: | Risk | Mitigation | |—| | | Neutrality perception (we compete + operate) | Independent board, one-harness rule, no special access, raw-data release (§L0) | | Honesty-charter breach | PBPP (§2); private suite never headlined | | Benchmark saturation / contamination | Hidden split + rotation + fresh scenarios (L1.d) | | Compute cost of running all systems | Budget realistically; curated reference set + community queue (Open-LLM-Leaderboard lesson) | | Over-claiming | Recall vs QA separated; CIs + ties; “provisional” until reproduced | | Depends on Plan A capabilities not ready | Part I can benchmark today’s system; publish deterministic strengths first, add QA as S1 lands |

6. Timeline, roles, dependency map

Sequencing (critical path bold): 1. **M4 charter** → **PBPP** (unblocks all publication) — days. 2. **M1 public harness** (Wave E kickoff) | **L0 governance charter** — weeks 1–3. 3. **M2/M3 bundle + third-party repro** | **L1 methodology + L2 site scaffold** — weeks 2–8. 4. L3 explainers | finalize Part I results — weeks 4–10. 5. **L4 launch** (seed ≥8 systems, methods paper) — weeks 8–14.

Dependencies: M4 blocks any publication; M1’s harness is L1’s engine (shared eval core — serialize writes); L0 blocks L4; Part I depends on **Plan A** capabilities for the QA numbers (deterministic numbers can publish independently first).

Roles: *eval-eng agent* (M1–M2), *web agent* (L2–L3), *governance/human operator* (L0 board, M3 third-party repro), *research/writing* (L0.4 methods paper). Independent workstreams run as parallel sub-agents; serialize shared-eval-core writes.

Definition-of-Done checklist (Plan B)

- ☐ Charter updated to PBPP; provider-bakeoff README references it (M4).
- ☐ `eval/public/` harness live; LongMemEval-recall, HippoRAG multi-hop, MemoryAgentBench adapter, BEAM runnable with bundles (M1).
- ☐ Reproducibility bundle standard implemented; agent-reproduces from bundle (M2).
- ☐ Independent third-party reproduction of headline numbers on file (M3).
- ☐ Governance board seated; COI + firewall policy public (L0).
- ☐ Methodology + anti-gaming + contamination controls implemented (L1).
- ☐ Leaderboard site live from a public, versioned results repo; trace browser working (L2).
- ☐ Explainer/education layer published (L3).
- ☐ Public launch: ≥8 systems, methods paper, raw-data dump openly licensed (L4).
- ☐ Mnemosyne entered under identical rules; no special access.

Appendix A — Dependency on Plan A

Plan A (*The Memory System*) produces the capabilities this plan measures: elite retrieval, multi-hop synthesis (S1, the QA-number driver), security-under-attack and calibration (S3), and scale numbers (S4). Deterministic-retrieval results can be benchmarked and published on today’s system; QA numbers improve as Plan A S1 lands.

Appendix B — Leaderboard-credibility source map

MTEB (arXiv 2506.21182; docs.mteb.org), HELM (2211.09110; crfm.stanford.edu), Chatbot Arena / LMArena (2403.04132) + Leaderboard Illusion (2504.20879) + Arena response (arena.ai/blog/our-response), GLUE/SuperGLUE (1804.07461 / 1905.00537), SWE-bench Verified (openai.com/index/introducing-swe-bench-verified) + SWE-bench Illusion (2506.12286), ARC-AGI (arcprize.org), Kaggle (kaggle.com/docs/competitions), Papers-with-Code shutdown (github.com/paperswithcode/paperswithcode-data/issues/116), memory-leaderboard call (2603.07670), in-repo landscape (docs/research/AI-Memory-Systems-Market-Research-2026.md).

This plan changes posture deliberately (public benchmarks + a public leaderboard) but only through PBPP, which is stricter than the field norm. Mnemosyne competes on the leaderboard under the same rules as every other system.